Employee Table Schema with Data Integrity Rules

Created By: Sharu latha. B

Course: AI-Driven Data Analytics
Batch No: FNB16

Project Objectives

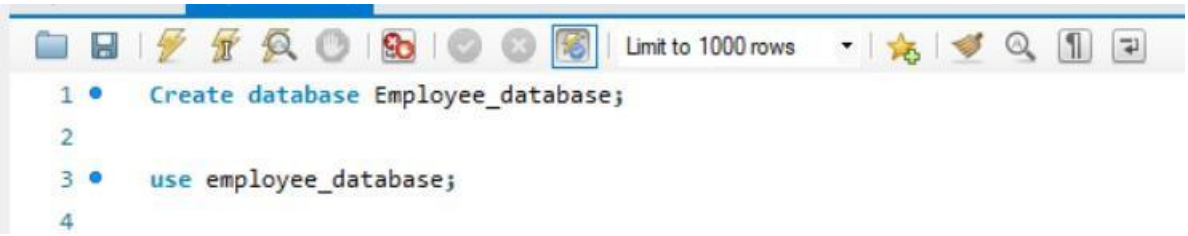
The objective of this project is to design and implement a relational database system for managing employee information. The project focuses on applying SQL constraints to ensure data integrity, consistency, and reliability across multiple tables.

MySQL DDL Commands

- Create
- Alter
- Drop
- Truncate
- Rename

Creating Employee Database with Constraints

Database and Table Creation (CREATE):



Importance of Creating Tables Inside a Database
Organized Data Storage

- ✦ Tables provide a structured way to store information in rows and columns.
- ✦ Each table represents a specific entity (e.g. Employees, Departments, Locations).

Data Integrity

- ✦ Constraints (PRIMARY KEY, FOREIGN KEY, NOT NULL, CHECK) ensure that only valid data is stored.
- ✦ Prevents duplication and enforces rules automatically.

Table Creation (CREATE):

Example:

Table: Departments
Attributes:
• Department id – Primary Key
• department name

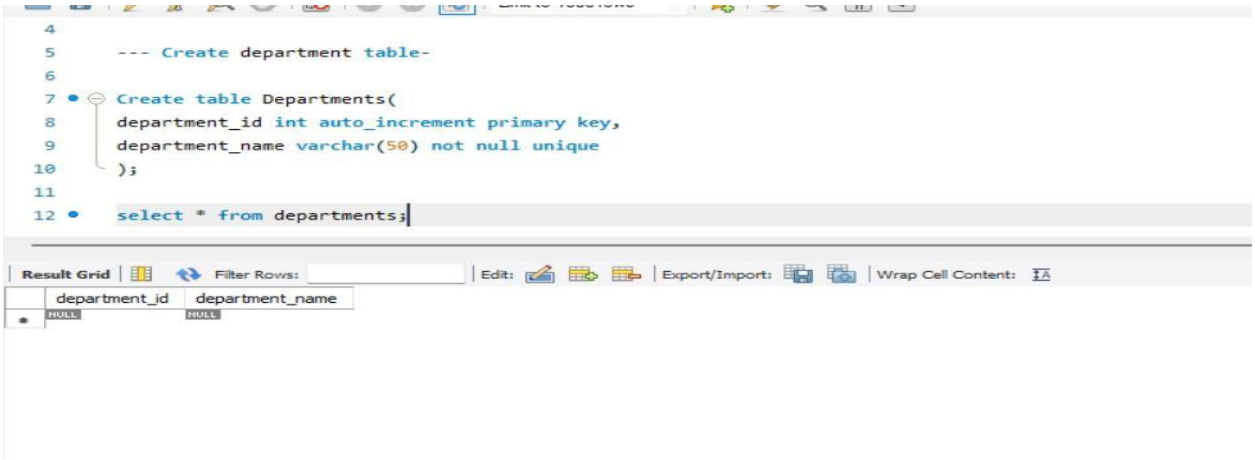


Table: Location
Attributes:
✦ location id – Primary Key
✦ location name

```
13
14      --- create table Location-
15
16  •   Create table Location(
17      location_id int auto_increment primary key,
18      location_name varchar(100) not null unique
19      );
20
21  •   desc Location;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
location_id	int	NO	PRI	NULL	auto_increment
location_name	varchar(100)	NO	UNI	NULL	

Table: Employees
Attributes:
✦ employee id – Primary Key
✦ Employee name
✦ Gender
✦ Age
✦ Hire date
✦ Designation
✦ Salary
✦ Department id – Foreign Key referencing department (department id)
✦ location id—Foreign Key referencing location (location id)

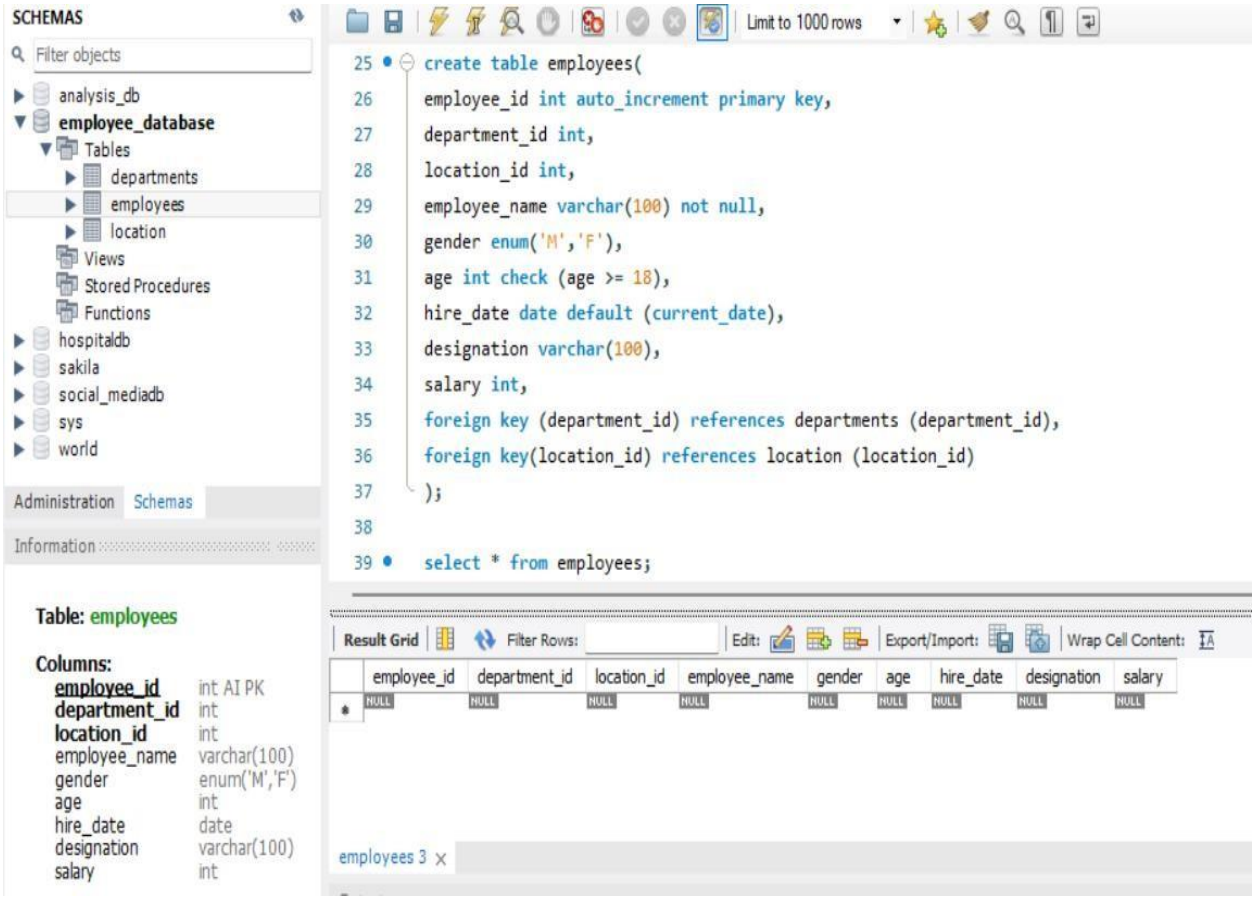


Table Alteration (ALTER)

- Add a new column named "email" to the Employees table to store employee email addresses.

```
39  select * from employees;
40  use employee_database;
41
42  --- Alter table-Add email column-
43  Alter table employees
44  add column Email varchar(100) unique;
45
46  select * from employees;
```

- **Modify the data type of the "designation" column in the Employees table to support a wider range of values.**

```
48      --- Modify the data type of the destination column- wider range of value--
49
50 •    Alter table employees
51      Modify designation varchar(400);
52
```

- **Drop the "age" column from the Employees table.**

```
54
55      --- drop age column  from employee table--
56
57 •    alter table employees
58      drop column age;
59
60
```

Table: employees

Columns:

employee_id	int AI PK
department_id	int
location_id	int
employee_name	varchar(100)
gender	enum('M','F')
hire_date	date
designation	varchar(400)
salary	int

Table Renaming (RENAME):

- **Rename the “hire date” column to “date of joining”.**

```
61
62 •    Alter table employees
63      rename column hire_date to date_of_joining;
64
65
66
```

Table: employees

Columns:

employee_id	int AI PK
department_id	int
location_id	int
employee_name	varchar(100)
gender	enum('M','F')
date_of_joining	date
designation	varchar(400)
salary	int

- **Rename the "Departments" table to "Departments Info".**
- **Rename the "Location" table to "Locations".**

```
55      --- Table Rename departments to department_info--
56
57 •    Rename table departments to department_info;
58
59 •    Rename table location to locations;
70
71
```

Table Truncation (TRUNCATE):

- **Truncate the Employees table**

```
71      --- Truncate employees table --
72
73 •    truncate table employees;
74
```

Database & Table Dropping (DROP):

```
78
79      /* Database and table dropping */
80
81 •    drop table employees;
82 •    drop database employee_database;
83
84
```

Database Recreation:

- **Recreate it using the provided schema, ensuring that all tables are created with the appropriate constraints as instructed.**

```
/* database recreation */

drop database employee_database;

Create database Employee_database;

use employee_database;
```

- **Establish links between the "department id" and "location id" fields in the "employees" table and their respective tables.**

```
--- create table employees-

create table employees(
employee_id int auto_increment primary key,
department_id int,
location_id int,
employee_name varchar(100) not null,
gender enum('M','F'),
age int check (age >= 18),
hire_date date default (current_date),
designation varchar(100),
salary int,
foreign key (department_id) references departments (department_id),
foreign key(location_id) references location (location_id)
);
```

Departments Table:

- **Ensure that the "department id" uniquely identifies each department.**

```
/*  Ensure that the "department_id" uniquely identifies each department */

 insert into departments(department_name)
 values('HR'),('Finance'),('IT'),('Sales');
```

```
125     /*  Ensure that the "department_id" uniquely identifies each department */
126
127     insert into departments(department_name)
128     values('HR'),('Finance'),('IT'),('Sales');
129
130     select * from departments;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

department_id	department_name
1	HR
2	Finance
3	IT
4	Sales
NULL	NULL

```
Create table Departments(
department_id int auto_increment primary key,
department_name varchar(50) not null unique
);
```

```
ALTER TABLE departments
ADD CONSTRAINT UNIQUE (department_id);
```

- **Set up constraints on the "department_name" to avoid duplicate and null entries.**

```
/* Set up constraints on the "department_name" to avoid duplicate and null entries */


SELECT *
FROM departments
WHERE department_name is not null;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

department_id	department_name
1	HR
2	Finance
3	IT
4	Sales
NULL	NULL

```
Create table Location(
location_id int auto_increment primary key,
location_name varchar(100) not null unique
);
```

Locations Table:

- **Establish a mechanism to automatically generate unique identifiers for each location, ensuring that they are incremented sequentially.**

```
/* Locations Table
• Establish a mechanism to automatically generate unique identifiers for each
location, ensuring that they are incremented sequentially */

insert into location( location_name)
values('Chennai'),('Coimbatore'),('Mumbai'),('Bangalore');

select * from location;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

location_id	location_name
1	Chennai
2	Coimbatore
3	Mumbai
4	Bangalore
NULL	NULL

- **Implement constraints to prevent the insertion of null and duplicate locations.**

```
148    /* Implement constraints to prevent the insertion of null and duplicate locations */
149
150 •  SELECT *
151    FROM location
152    WHERE location_name is not null;
153
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

location_id	location_name
1	Chennai
2	Coimbatore
3	Mumbai
4	Bangalore
NULL	NULL

Employees Table:

- **Guarantee that each employee has a distinct identifier.**

```
158  /* Employees Table:
159     Guarantee that each employee has a distinct identifier */
160
161 • INSERT INTO employees (employee_name, gender, age, hire_date, designation, salary,department_id, location_id)
162   VALUES ('Arun Kumar', 'M', 28, '2025-01-01', 'developer', 45000,1,1),
163   ('Karthik', 'M', 35, '2025-01-20', 'Finance Officer', 55000, 2, 2);
164
165 •   select * from employees;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	department_id	location_id	employee_name	gender	age	hire_date	designation	salary
1	1	1	Arun Kumar	M	28	2025-01-01	developer	45000
2	2	2	Karthik	M	35	2025-01-20	Finance Officer	55000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- **Create a restriction to ensure that the employee's name is always provided.**

```
167    /* Create a restriction to ensure that the employee's name is always provided */
168
169    select * from employees
170    WHERE employee_name is not null;
171
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	department_id	location_id	employee_name	gender	age	hire_date	designation	salary
1	1	1	Arun Kumar	M	28	2025-01-01	developer	45000
2	2	2	Karthik	M	35	2025-01-20	Finance Officer	55000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- **Limit the acceptable values for the "gender" field to only 'M' or 'F'.**

```
173    /* Limit the acceptable values for the "gender" field to only 'M' or 'F' */
174
175    select * from employees
176    WHERE gender not in ('M','F');
177
```

- **Enforce a condition to ensure that the employee's age is 18 or above.**

```
186
187
188    select * from employees
189    where age >=18;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	department_id	location_id	employee_name	gender	age	hire_date	designation	salary
1	1	1	Arun Kumar	M	28	2025-01-01	developer	45000
2	2	2	Karthik	M	35	2025-01-20	Finance Officer	55000
3	3	3	Rahul	M	18	2026-07-27	Intern	10000
4	4	4	Meena	F	20	2026-07-27	Software Tester	15000
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

employees 14 ×

- **Automatically assign the current date to the "hire_date" field if not specified.**

Edit: | Export/Import: | Wrap Cell Cor

location_id	employee_name	gender	age	hire_date	designation
1	Arun Kumar	M	28	2025-01-01	developer
2	Karthik	M	35	2025-01-20	Finance Officer
3	Rahul	M	18	2026-07-27	Intern
4	Meena	F	20	2026-07-27	Software Tester

Conclusion

- **The Employees Database was successfully created using SQL DDL commands.**
- **Tables for Employees, Departments, and Locations were designed with proper structure.**
- **Constraints such as PRIMARY KEY, FOREIGN KEY, NOT NULL, CHECK, and DEFAULT were applied to ensure data integrity.**
- **The hire_date field was set to automatically assign the current date if not specified.**
- **Sample data was inserted and tested against constraints to validate correctness.**
- **Invalid entries (like age below 18 or wrong gender values) were prevented by constraints.**
- **Relationships between tables were established using foreign keys, ensuring consistency.**

Thank You

Entri Elevate